

VRRP (Virtual Router Redundancy Protocol):

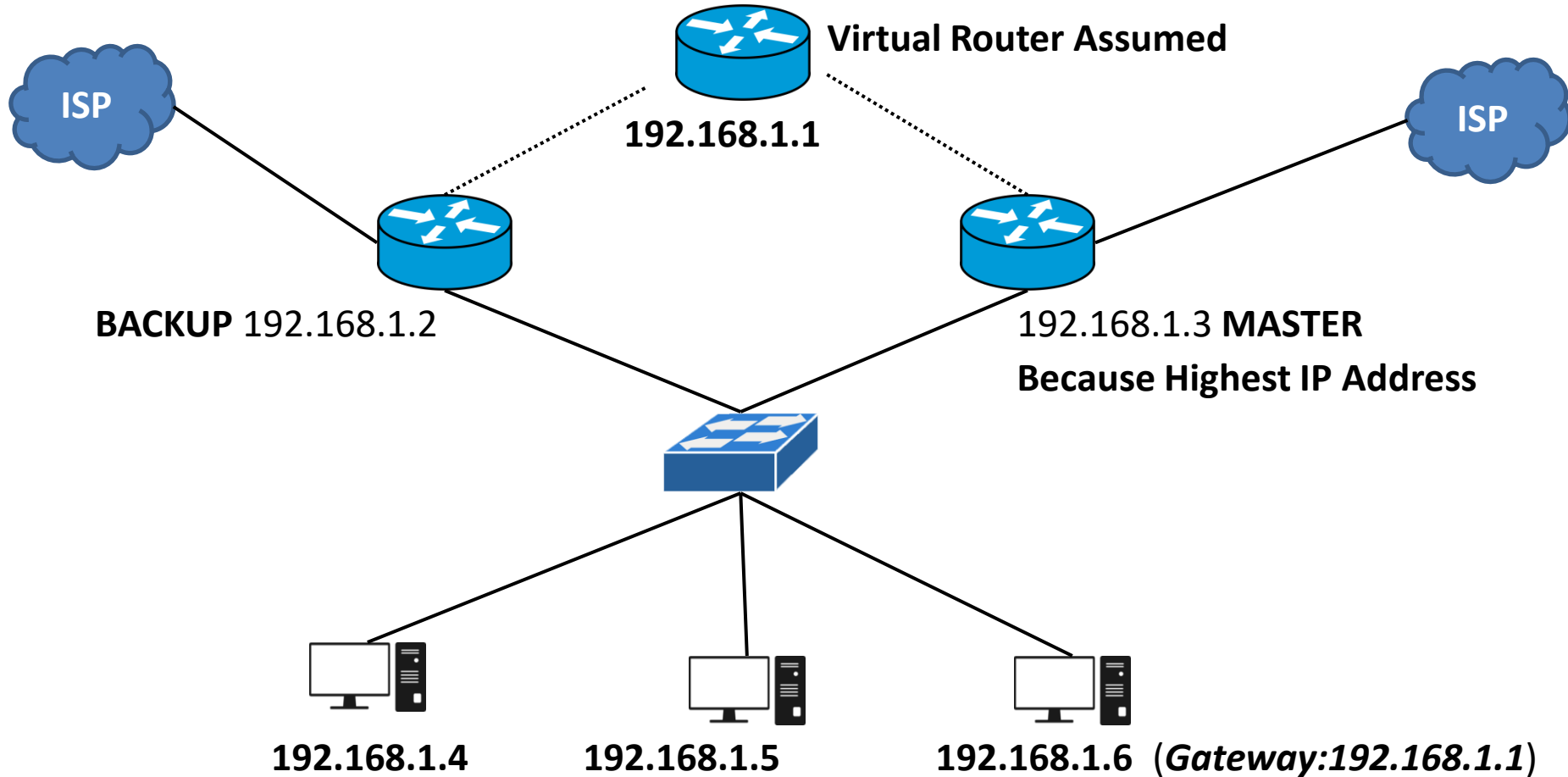
VRRP (Virtual Router Redundancy Protocol) is a network protocol that provides automatic assignment of available routers to participating hosts. It helps ensure high availability and reliability for a default gateway (router) by automatically failing over to a backup router if the primary one becomes unavailable.

Advertisement Timers:

Master routers send advertisements periodically (default: 1 second).

Backup routers listen for these advertisements and take over if they miss three consecutive messages.

VRRP :Virtual Router Redundancy Protocol:



VRRP State Transitions Overview:

Initialize → Backup: When the router is not the highest priority and is waiting to monitor the Master.

Initialize → Master: When the router has the highest priority and is ready to take control.

Backup → Master: When the current Master fails or a higher-priority Backup router preempts the Master role.

Master → Backup: When a higher-priority router preempts or takes over as the new Master.

VRRP Configuration: -

```
Router(Config)#interface GigabitEthernet0/1
```

```
Router(Config)# ip address 192.168.1.2 255.255.255.0
```

```
Router(Config)# vrrp 1 ip 192.168.1.1 (virtual Router)
```

```
Router(Config)# vrrp 1 priority 110
```

```
Router(Config)# vrrp 1 preempt
```

```
Router(Config)# vrrp 1 timers advertise 1
```

Explanation:

vrrp 1 ip 192.168.1.1: Configures virtual IP address for VRRP group 1.

VRRP 1 priority 110: Sets the router's VRRP priority to **110**.

standby 1 preempt: Enables preemption, allowing the router to take over as the Master router if its priority is higher than Master router.